

MARHAM, United Kingdom

WMO: 034820

Lat: 52.6514N

Lon: 0.5661E

Elev: 23

StdP: 101.05

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.6	(d) -6.1	(e) 2.3	(f) -2.0	(g) -4.7	(h) 2.6	(i) -1.0	(j) 14.8	(k) 9.6	(l) 13.4	(m) 9.4	(n) 2.0	(o) 20	(p) 0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	27.0	18.9	25.0	18.0	23.2	17.1	19.8	25.2	18.8	23.6	17.9	21.9	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.8	21.8	16.9	12.1	20.6	16.2	11.5	19.8	56.6	25.3	53.3	23.7	50.4	22.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.2	(b) 10.9	(c) 9.6	(e) -7.4	(f) 30.8	(g) 2.4	(h) 1.6	(i) -9.1	(j) 31.9	(k) -10.5	(l) 32.9	(m) -11.9	(n) 33.8	(o) -13.6	(p) 35.0		
(a) 12.2	(b) 10.9	(c) 9.6	(e) -7.8	(f) 22.0	(g) 2.4	(h) 1.5	(i) -9.6	(j) 23.1	(k) -11.0	(l) 24.0	(m) -12.4	(n) 24.8	(o) -14.1	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	4.1	4.6	6.4	9.0	12.0	14.8	17.2	17.1	14.4	11.1	7.0	4.6
	DBStd	5.61	3.26	3.28	3.06	3.10	3.12	2.82	2.68	2.65	2.58	3.05	3.17	3.63
	HDD10.0	817	183	153	117	55	16	1	0	0	1	23	99	171
	HDD18.3	3007	440	384	370	281	197	111	56	57	120	224	341	427
	CDD10.0	899	1	2	5	24	77	146	223	220	134	57	8	2
	CDD18.3	48	0	0	0	0	1	6	20	19	3	1	0	0
	CDH23.3	383	0	0	0	2	9	52	164	132	21	3	0	0
	CDH26.7	72	0	0	0	0	0	9	31	29	2	1	0	0
Wind	WSAvg	4.6	5.2	5.2	5.0	4.6	4.6	4.2	4.1	4.1	4.2	4.7	4.7	4.9
Precipitation	PrecAvg	675	58	46	41	44	50	56	63	67	58	65	67	60
	PrecMax	842	99	87	83	138	124	154	114	156	109	118	113	110
	PrecMin	477	16	13	6	1	12	16	27	6	15	17	20	28
	PrecStd	107	23	19	21	32	27	37	26	35	26	25	23	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.4	16.7	22.1	24.6	28.0	29.9	29.6	25.9	21.1	15.8	13.7
		MCWB	11.2	11.2	11.3	14.4	16.8	19.8	20.0	19.6	18.3	16.2	14.0	11.9
	2%	DB	11.6	12.2	14.7	18.5	22.2	24.6	27.1	26.8	22.9	18.5	14.2	12.5
		MCWB	10.1	9.9	10.8	12.7	15.9	17.8	18.9	19.0	17.1	15.1	12.5	11.1
	5%	DB	10.3	10.8	13.1	16.3	20.2	22.4	25.1	24.3	20.9	17.0	13.0	11.2
		MCWB	9.0	8.9	9.9	11.4	14.7	16.7	18.0	17.7	16.1	14.4	11.4	10.0
	10%	DB	9.1	9.6	11.6	14.6	18.0	20.5	23.1	22.4	19.2	15.7	11.8	10.0
		MCWB	7.9	8.0	9.0	10.4	13.4	15.5	17.1	16.9	15.1	13.6	10.5	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.7	12.6	15.2	18.3	20.8	21.1	21.0	19.6	17.1	14.3	12.3
		MCDB	12.6	13.5	15.3	20.4	22.8	26.9	27.7	27.9	24.0	20.1	15.4	13.3
	2%	WB	10.3	10.2	11.4	13.2	16.4	18.5	19.8	19.6	17.9	15.8	12.8	11.3
		MCDB	11.4	11.9	13.9	17.7	20.9	22.8	25.5	24.6	21.5	17.9	13.9	12.3
	5%	WB	9.1	9.2	10.3	12.1	15.2	17.3	18.7	18.6	16.8	14.9	11.7	10.1
		MCDB	10.2	10.5	12.6	15.5	19.4	21.2	23.8	23.2	19.8	16.5	12.8	11.2
	10%	WB	8.0	8.2	9.3	10.9	13.9	16.2	17.7	17.6	15.9	14.0	10.6	9.0
		MCDB	9.0	9.6	11.4	14.0	17.5	19.7	22.0	21.3	18.3	15.5	11.6	10.0
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.2	7.6	8.9	9.1	8.9	9.4	9.3	8.8	7.4	6.1	5.3
		MCDBR	6.3	7.3	9.7	11.4	12.3	12.0	12.7	12.0	10.8	7.9	6.3	5.7
		MCWBR	5.8	5.5	6.4	6.6	7.1	6.4	5.8	5.3	5.8	5.3	5.3	5.3
	5% WB	MCDBR	6.4	6.8	8.9	10.7	11.4	10.6	11.6	10.8	9.6	7.2	6.1	5.7
		MCWBR	6.0	5.5	6.4	6.5	7.0	6.2	5.9	5.3	5.9	5.4	5.4	5.4
Clear-Sky Solar Irradiance	taub	0.326	0.343	0.377	0.402	0.405	0.410	0.410	0.398	0.377	0.357	0.330	0.316	
	taud	2.353	2.374	2.274	2.222	2.263	2.271	2.283	2.334	2.391	2.426	2.401	2.338	
	Ebn at Noon	655	762	803	828	846	843	836	825	799	736	647	594	
	Edh at Noon	61	79	106	126	127	127	123	111	94	75	58	52	
All-Sky Solar Radiation	RadAvg	0.71	1.40	2.53	3.98	4.86	5.12	4.91	4.14	3.08	1.78	0.89	0.57	
	RadStd	0.07	0.20	0.27	0.41	0.37	0.39	0.50	0.29	0.26	0.15	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page